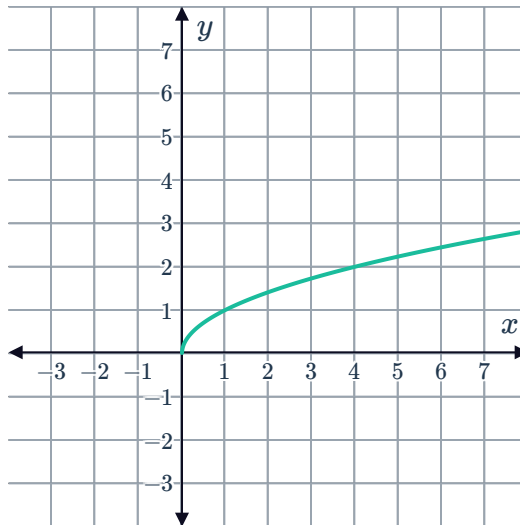


Graphs of square root functions



1

- a No
- b ∞
- c



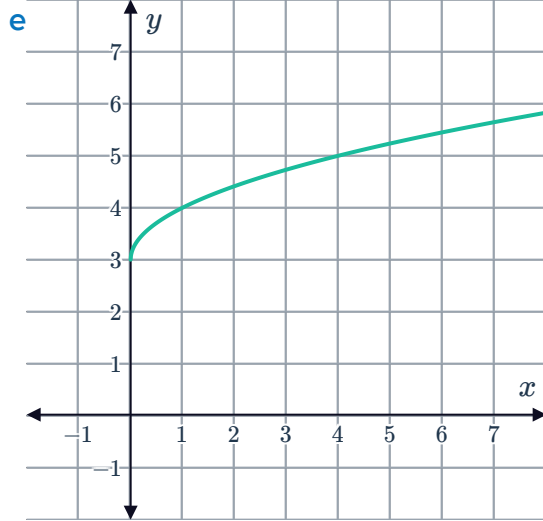
- d As x increases, the function increases at a slower and slower rate.
- e $(0,0)$
- f No
- g No
- h $y = 5\sqrt{x}$ increases more rapidly than $y = \sqrt{x}$.

2

- a No
- b The function decreases at a slower and slower rate.

3

- a No
- b ∞
- c $y = 3$
- d 0



4

a $x \leq 0$

b $y \geq 0$

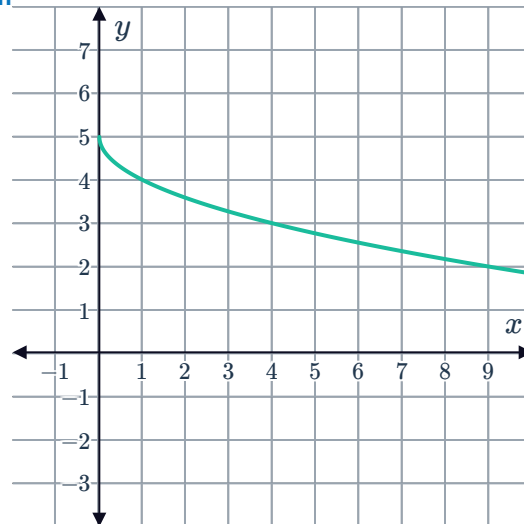
c ∞

5

a i $x \geq 0$

ii $y \leq 5$

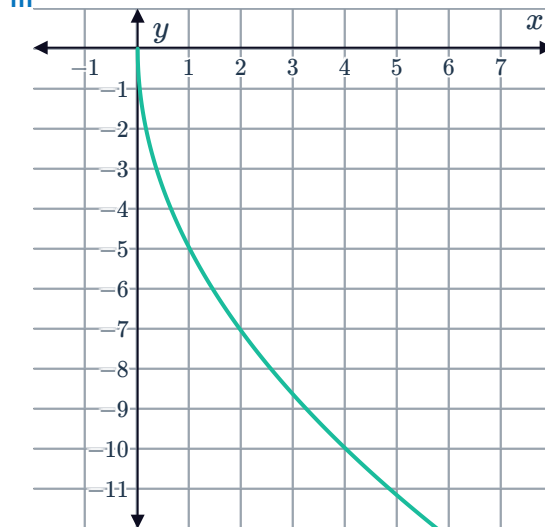
iii



b

i $x \geq 0$

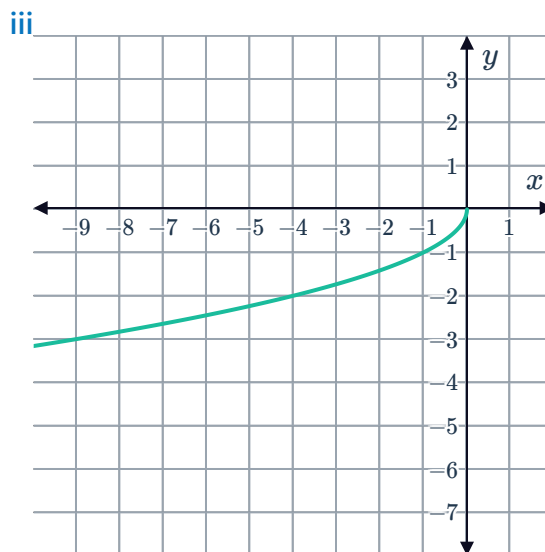
ii $y \leq 0$



c

i $x \leq 0$

ii $y \leq 0$



6

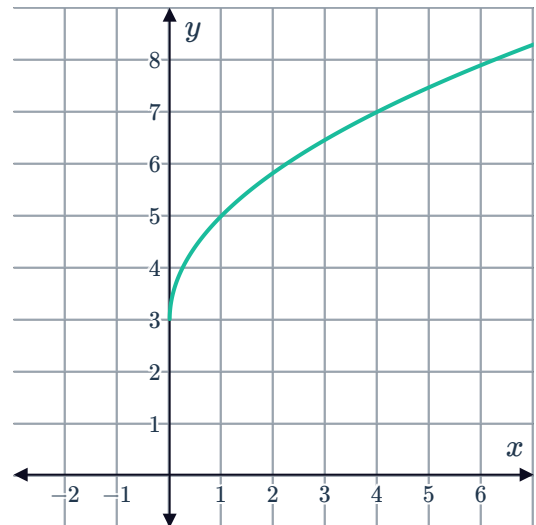
a Increasing

c $(0, 3)$



b More steep

d



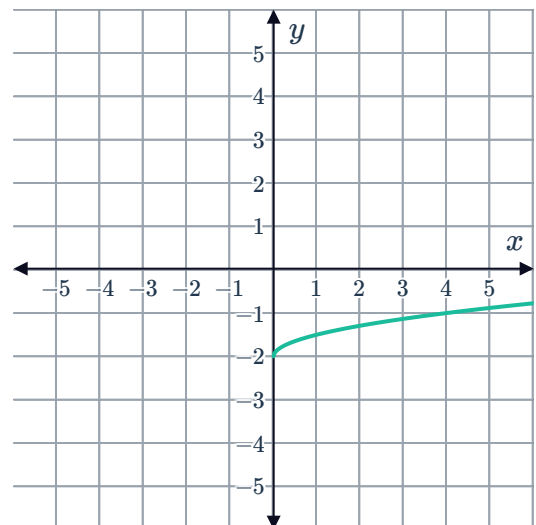
7

a Increasing

c $(0, -2)$

b Less steep

d



8

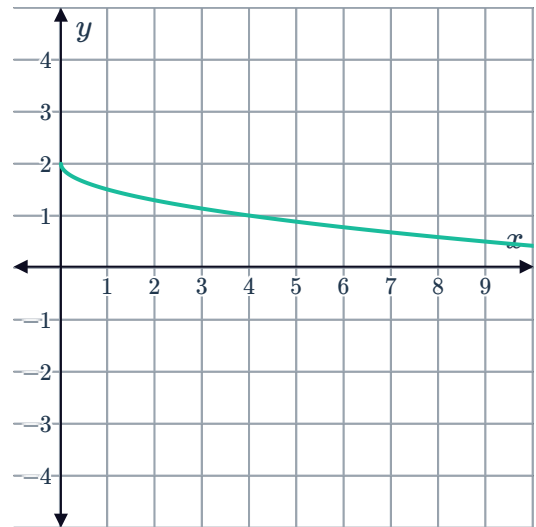
a Decreasing

c $(0, 2)$



b Less steep

d



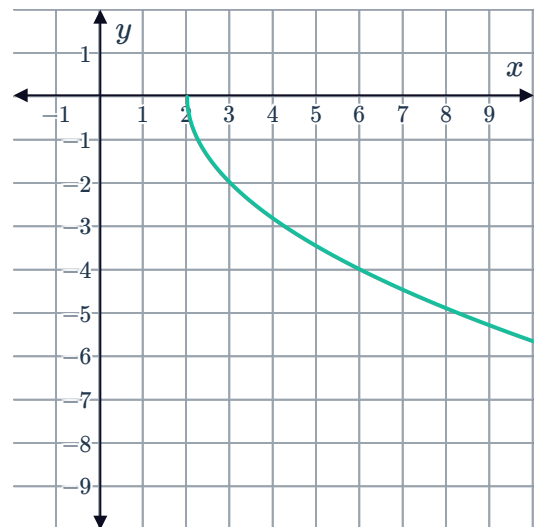
9

a Decreasing

c $(2, 0)$

b More steep

d



10

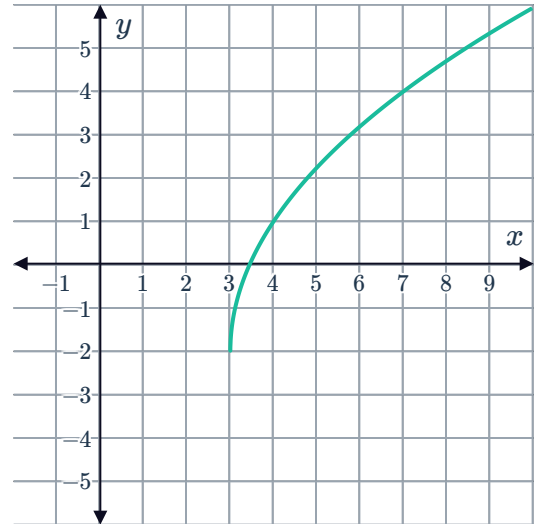
a Increasing

c (3, 2)



b More steep

d



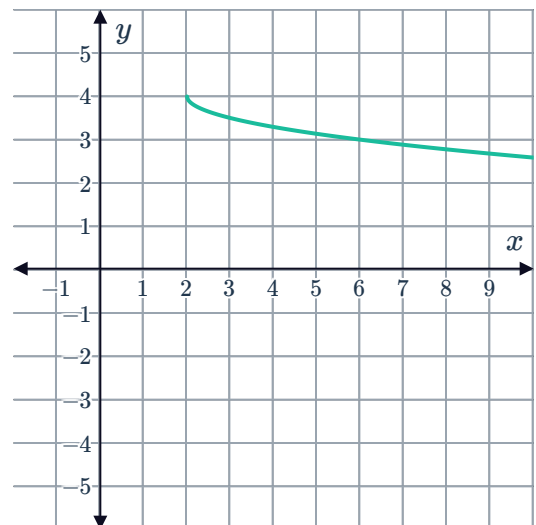
11

a Decreasing

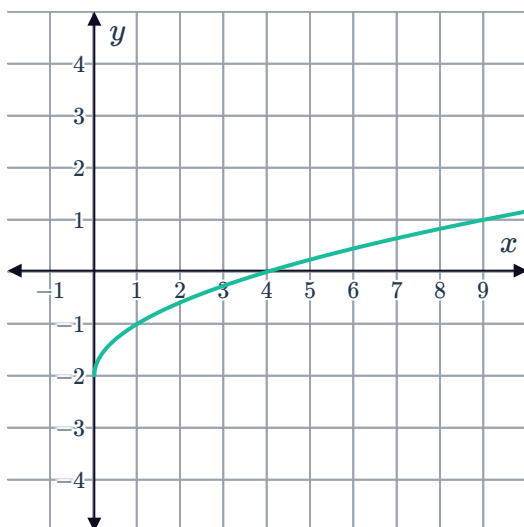
c (2, 4)

b Less steep

d



i

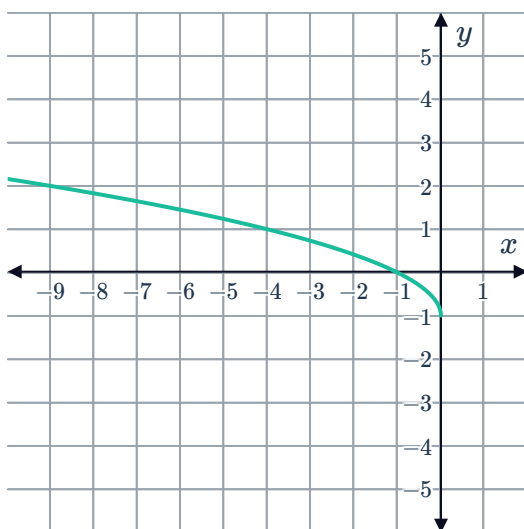


ii $x \geq 0$

iii $y \geq -2$

c

i



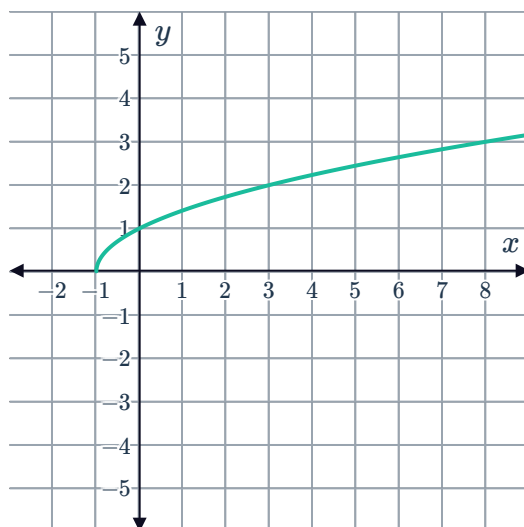
ii $x \leq 0$

iii $f(x) \geq -1$



b

i

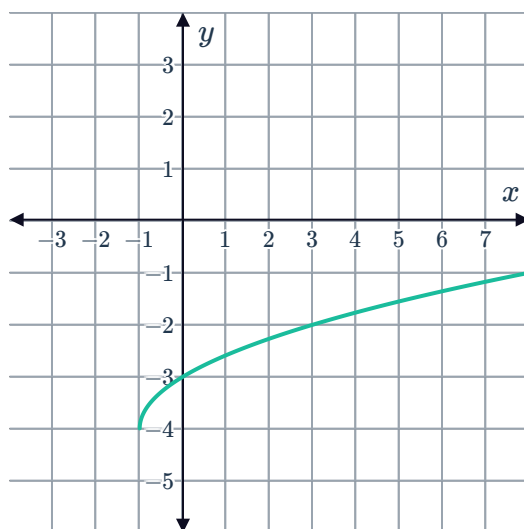


ii $x \geq -1$

iii $y \geq 0$

d

i

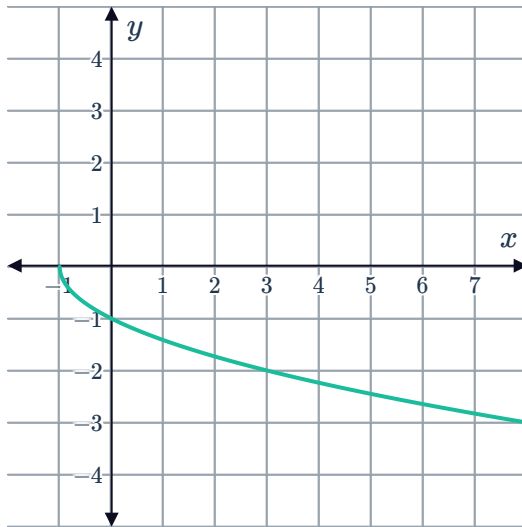


ii $x \geq -1$

iii $y \geq -4$

e

i

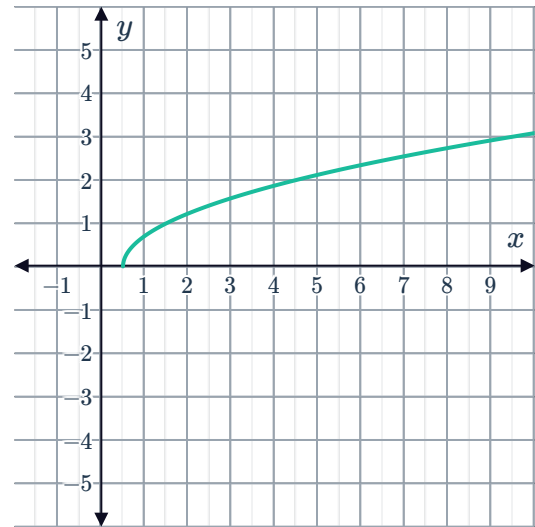


ii $x \geq -1$

iii $y \leq 0$



i

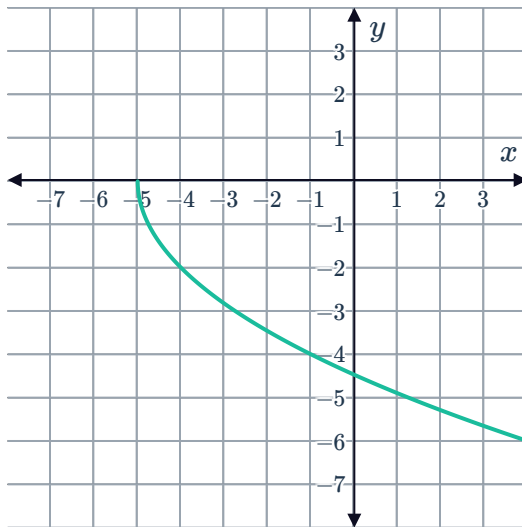


ii $x \geq 2$

iii $y \geq 0$

g

i

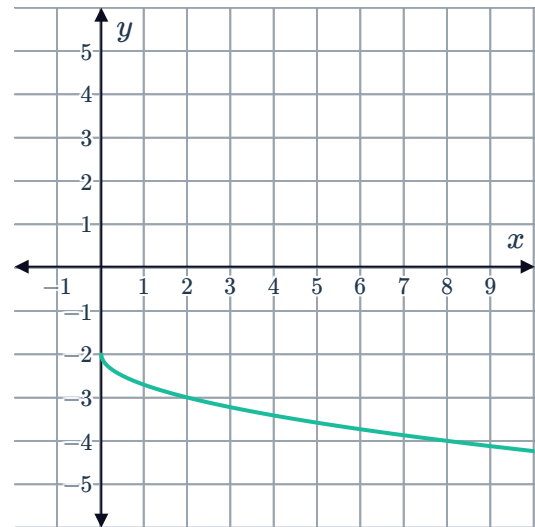


ii $x \geq -5$

iii $y \leq 0$

h

i

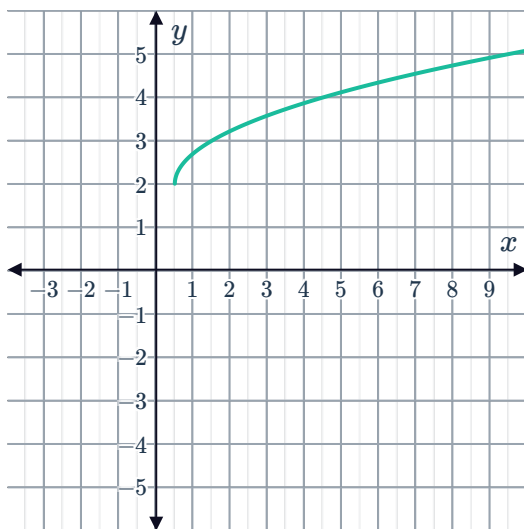


ii $x \geq 0$

iii $y \leq -2$

i

i



ii $x \geq 1$

iii $y \geq 2$

Characteristics of square root functions

13

a No

b No

c No

d No

e Yes

14 Domain: $x \geq 0$, range: $y \geq -2$

15

a $x \geq 5$

b $y \geq 0$

c

i Yes

ii No

iii No

16

a 6

b $x \leq 0$

c $y \geq 6$

17 $[2, \infty)$



Transformations

18

a $y = 2$

c $f(x) = \sqrt{x} + 2$

b Vertical shift 2 units up

19

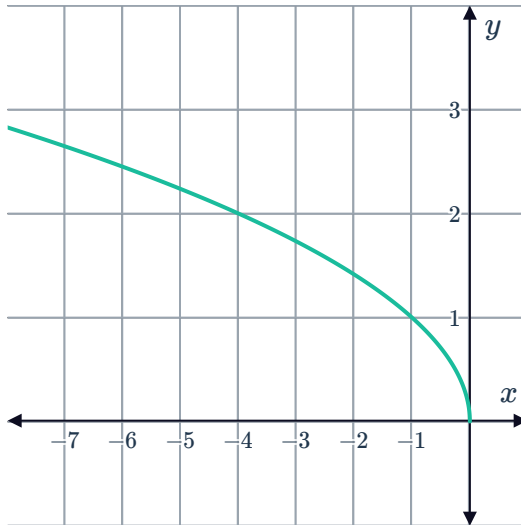
a $x = -2$

c $f(x) = \sqrt{x+2}$

b Horizontal shift 2 units to the left

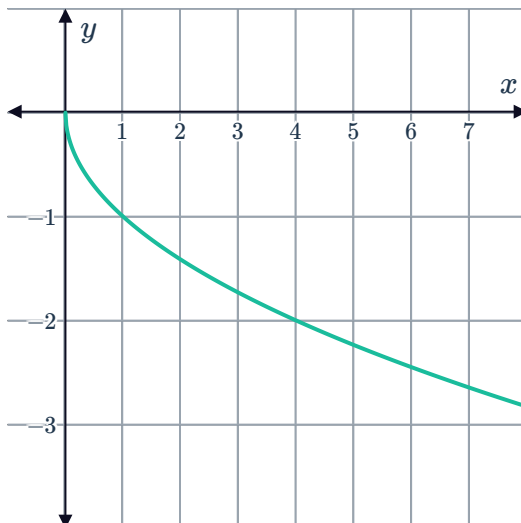
20

a
i



ii $y = \sqrt{-x}$

b
i



ii $y = -\sqrt{x}$

21

a $x \leq 0$



b $y = 3\sqrt{-x}$

22

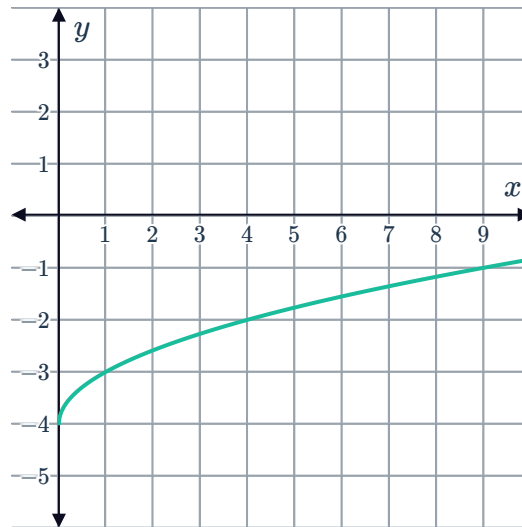
a 3

b Downwards

23

a The original function was translated down 4 units.

b



24 y increases more rapidly on $y = 3\sqrt{x}$ than on $y = \sqrt{x}$

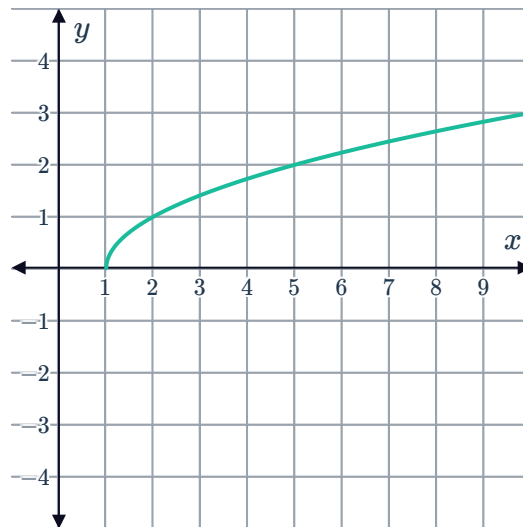
25

a $(0, -2)$

b $(0, 3)$

26

a



b Horizontal translation 1 unit(s) to the right

27

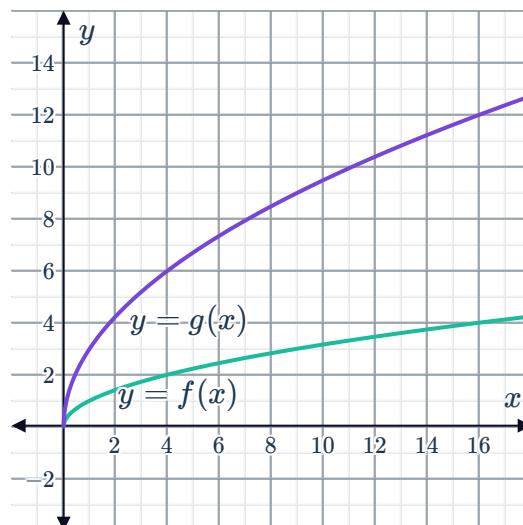
- Reflection across x -axis, then a translation 5 units down
- Translation 5 units up, then a reflection across x -axis

28

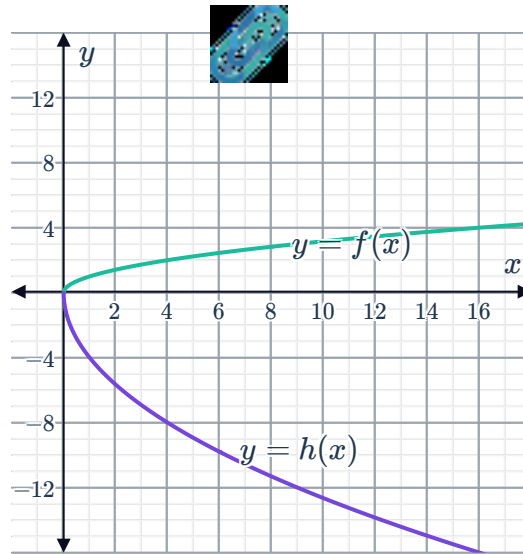
a

x	0	1	4	9	16
$y = f(x)$	0	1	2	3	4
$g(x) = 3f(x)$	0	3	6	9	12
$h(x) = -4f(x)$	0	-4	-8	-12	-16

b



d

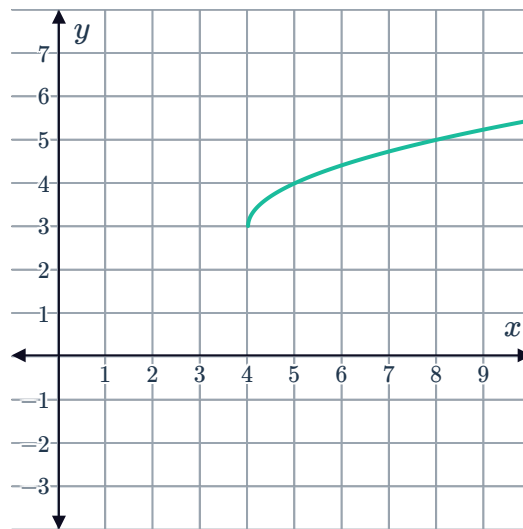


e Vertically stretched by a factor of 4 and reflected across the x -axis

29

a Translate the graph 4 units to the right and 3 units up.

b



30

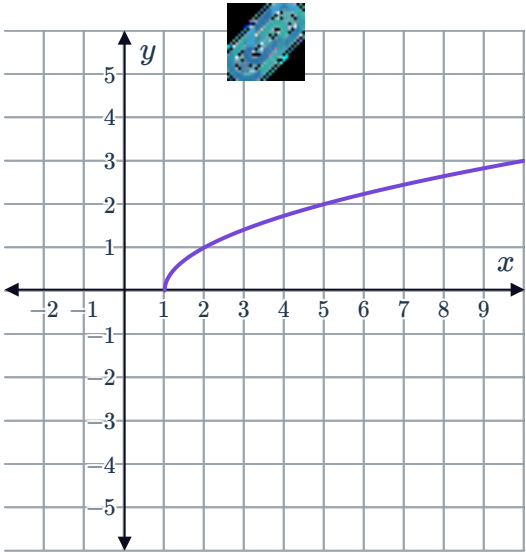
a $y = \sqrt{x-5}$

b $y = -\sqrt{x} + 6$

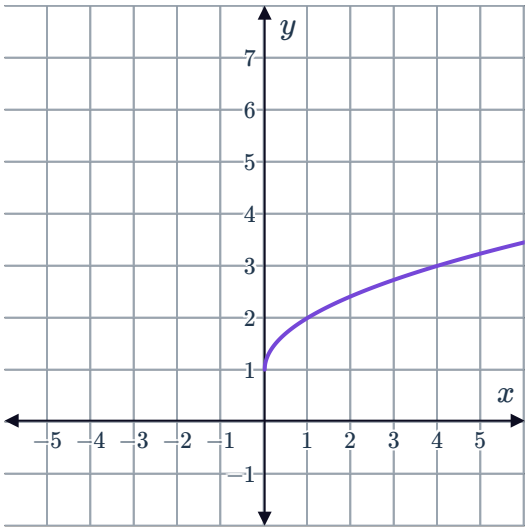
c $y = -\frac{1}{2}\sqrt{x}$

31 $\frac{\sqrt{2x}}{4} + 0$

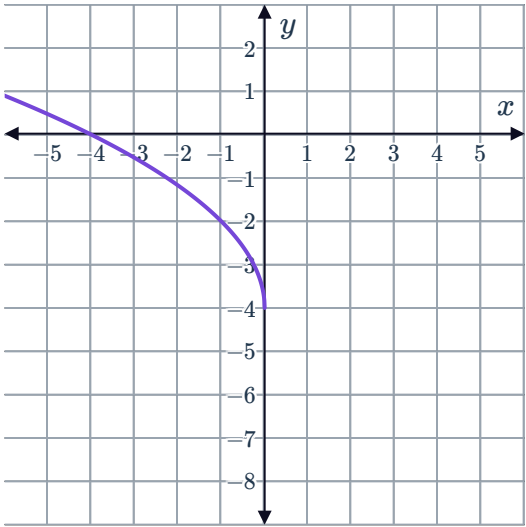
32



33



34





Applications

35 a $4\sqrt{5}$ miles

b 8.9 miles

36 1.57 seconds

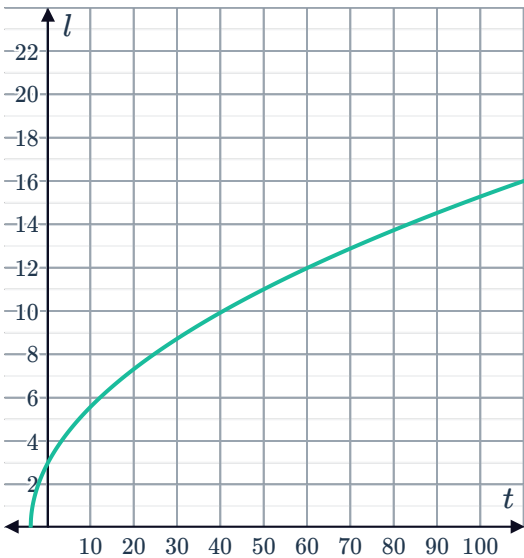
37 a $12\sqrt{2}$ feet per second

b 17 feet per second

38 a

Months (t)	0	60	96
Length (l)	3	12	15

b



c $t \geq 0$

d No

e 14.1 meters

f 0.7

39 $A = 14657415$

40

a $A = 2w^2$



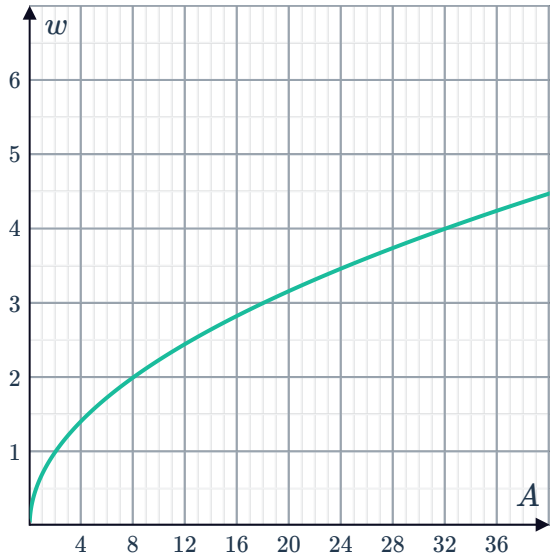
b

$$w = \sqrt{\frac{A}{2}}$$

c

Area (A)	0	2	8	18	32
Width (w)	0	1	2	3	4

d



e 50 units²

f Above

41

a $D = -\sqrt{10} \text{ t}$

b $-5\sqrt{2} \text{ meters}$

c $\sqrt{2} \text{ m/s}$

d $2 - \sqrt{2} \text{ m/s}$